25901884 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Value Addition Course(VAC) - MG3VACBCH200 - MICROPLASTICS AND ENVIRONMENT

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which statement best explains why rivers are important in the transport of microplastics? [U] / [CO1]

   a). They increase the salinity of oceans

   b). They act as natural filters

   c). They carry microplastics from land to oceans

   d). They dissolve microplastics before reaching oceans

2. Which best explains why microplastics undergo biomagnification? [U] / [CO1]

   a). They dissolve easily in water

   b). They are converted to nutrients

   c). They are excreted rapidly by organisms

   d). They accumulate in tissues and are transferred through the food chain

3. The impact on soil fauna by the ingestion of microplastics are [U] / [CO2]

   a). Increased metabolic energy

   b). Rapid molting cycles

c). Physical harm, reduced growth d). None of these

4. Indicate the reason why it is difficult to avoid microplastic contamination in foods like honey and sugar. [U] / [CO3]

   a). There are no regulations about plastic in food

   b). Honey and sugar naturally attract plastic particles from air

   c). Micrcoplastics are widespread in the environment and can enter food at many stages.

   d). None of the above

5. Which natural system is disturbed by the presence of microplastics in oceans? [K] / [CO4]

   a).Nitrogen cycle b).Biological carbon pump

   c).Water cycle d).Ozone layer

[1x5 = 5]

Part B

Fill In The Blank Questions

Answer all questions

Each question carries **2** marks

6. The type of radiation that degrades plastics is _____ [K] / [CO1]

7. Microplastics that result from the breakdown of larger plastic items are known as _____ microplastics. [U] / [CO1]

8. Economic losses in _____ are linked to polluted beaches and degraded marine environments. [K] / [CO2]

9. Microplastics in the digestive tract can cause _____ of the gastrointestinal lining. [K] / [CO3]

10. Guidelines for industries are created to reduce _____ and promote _____ management. [K] / [CO4]

[2x5 = 10]

Part C

Short Answer Questions

Answer any **5** questions

Each question carries **3** marks

11. Define microplastic degradation. Give two examples of environmental factors that influence the degradation. [K] / [CO1]

12. Define bioaccumulation and explain the difference between bioaccumulation and biomagnification. [K] / [CO1]

13. Give an outline regarding the sources of Microbeads. [U] / [CO1]

14. Label the microplastic exposure pathways for fish and shellfish, and explain the observed effects. [K] / [CO2]

15. Why is bottled water often found to contain higher levels of microplastics than tap water? [An] / [CO3]

16. Explain the impacts of microplastics on the cryosphere. How is albedo influenced by microplastics? [U] / [CO4]

17. Recall the effects of microplastics on habitat destruction. [K] / [CO3]

[3x5 = 15]

Part D

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

18. Explain how sewage sludge and biosolids used as fertilizers contribute to soil microplastic pollution. [K] / [CO1]

19. Recall the role of industrial abrasives and pre-production pellets as microplastics. [U] / [CO1]

20. Relate the key effects of microplastics on freshwater ecosystems to broader ecological and human impacts. [U] / [CO2]

21. Describe how scientific limitations contribute to uncertainty in microplastic health research. [U] / [CO3]

22. Outline primary exposure route of microplastics to humans [U] / [CO1]

23. Compare the physical and biological methods of microplastic removal strategy. [U] / [CO4]

[5x4 = 20]